BLOC-BOT

TODO: author

October 5, 2026

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Chapter 1

Motor bracket design

The motor and sensor dimensions are taken from the reference models [1], [2], and the bracket was modelled in Autodesk Fusion [3]. Figure 1.4 shows the encoder disc in the sensor fork.

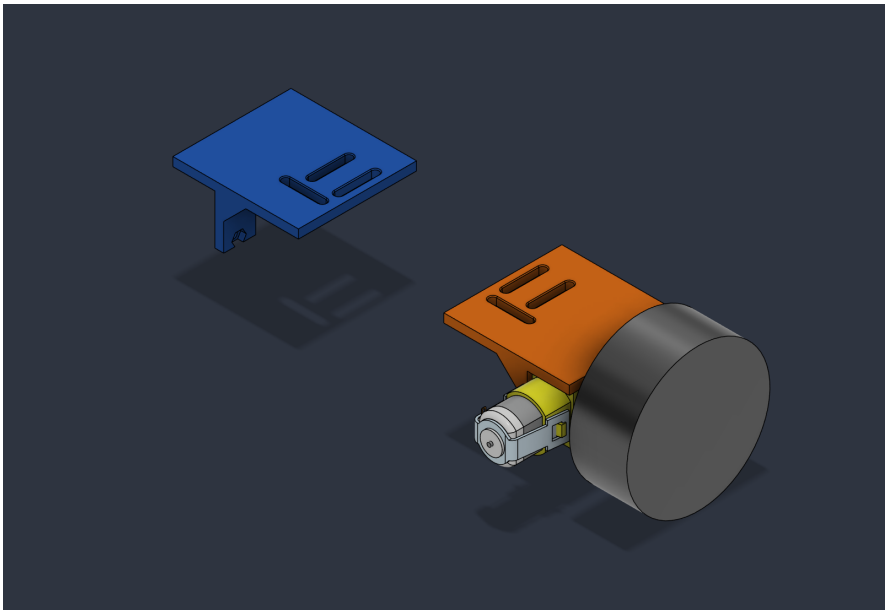


Figure 1.1: Right and left motor brackets with the right-hand TT motor and wheel (isometric view).

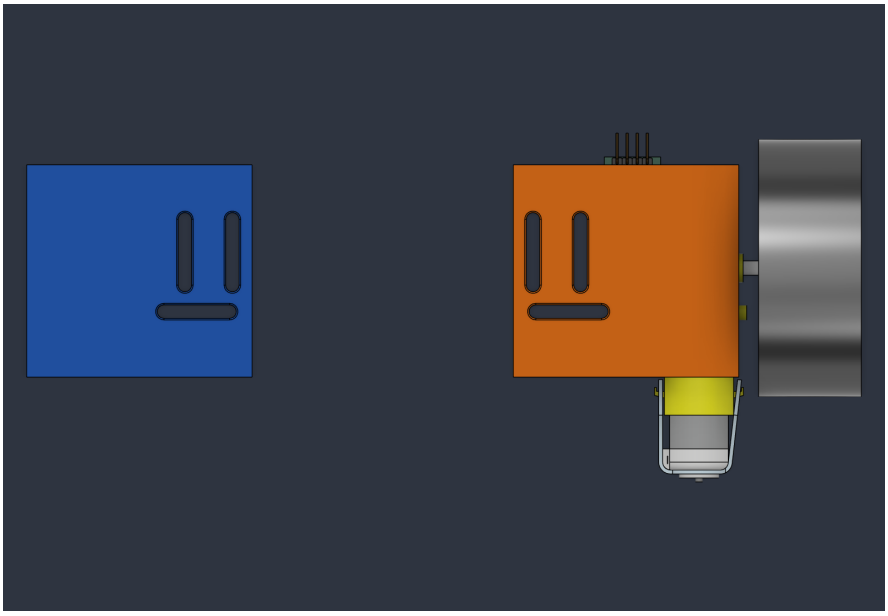


Figure 1.2: Top view of both brackets showing the three mounting slots.

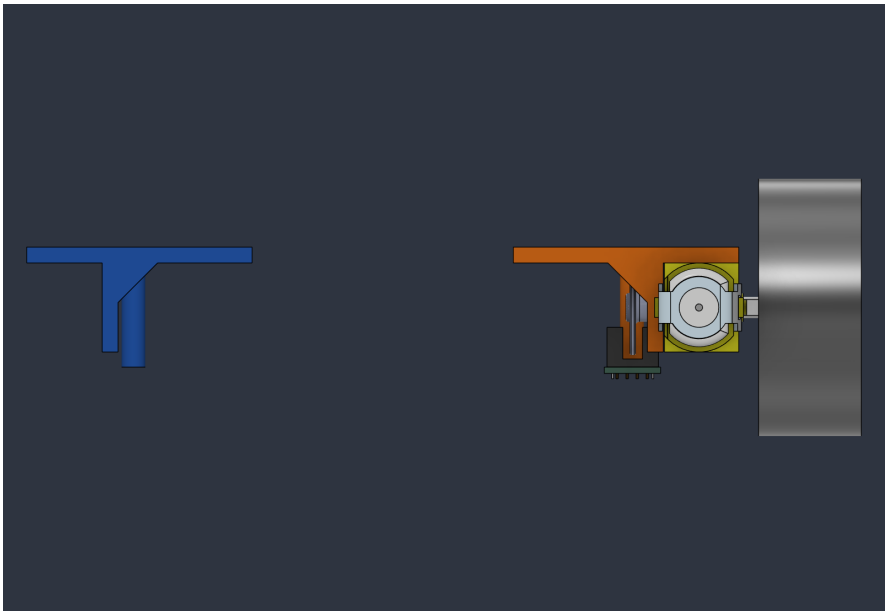


Figure 1.3: Front view: flange on the gearbox top, motor tab and sensor column.

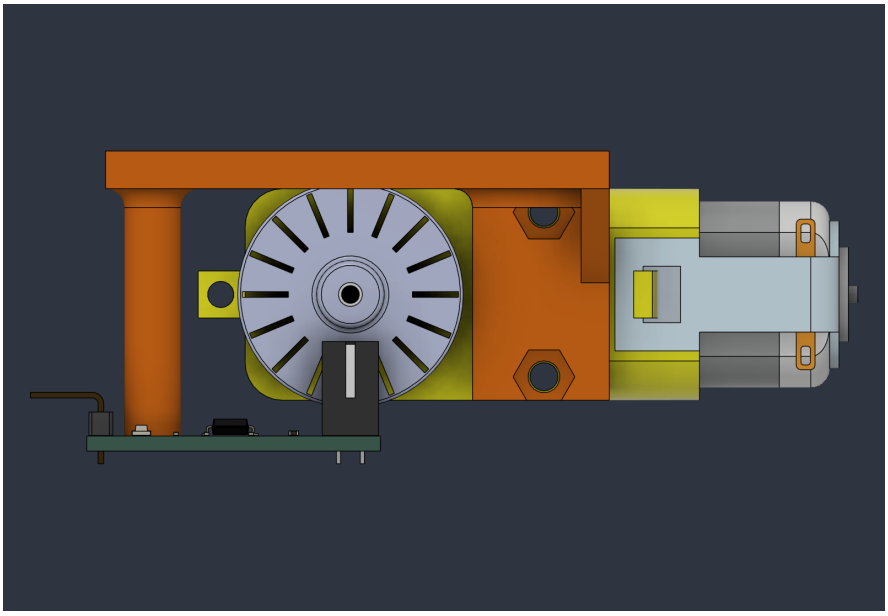


Figure 1.4: Side view: the encoder disc's bottom rim sits in the LM393 fork; the board is screwed to the column.

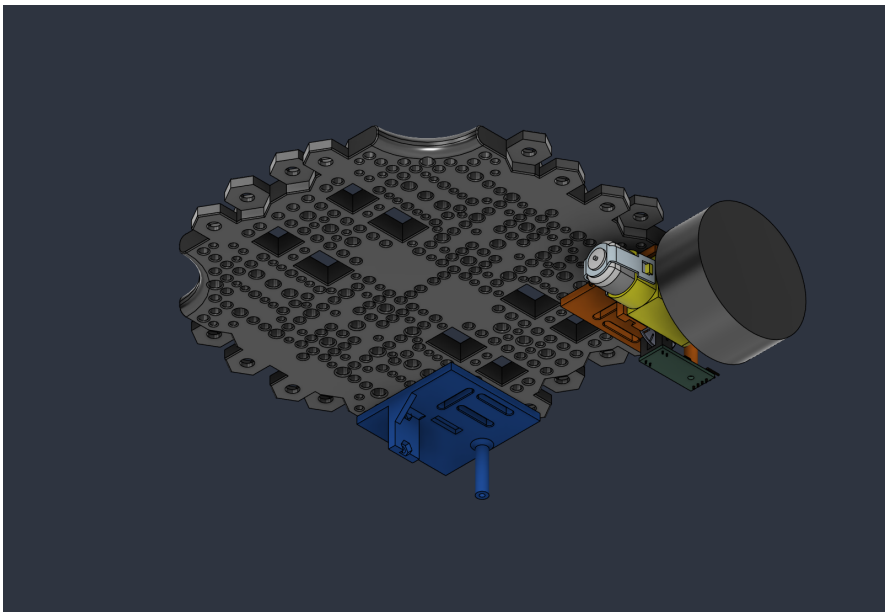


Figure 1.5: Brackets mounted under the waffle plate, seen from below.

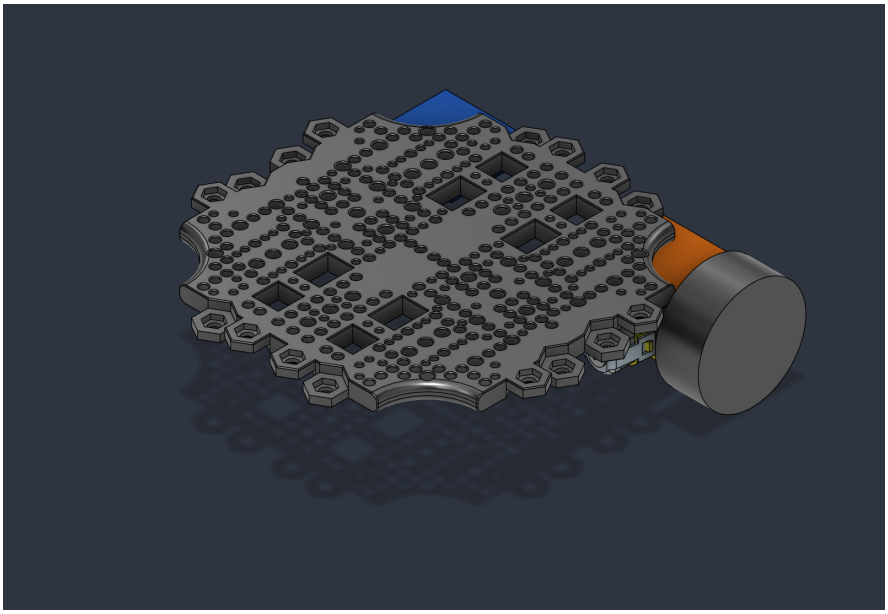


Figure 1.6: Brackets and wheel relative to the waffle plate, seen from above.

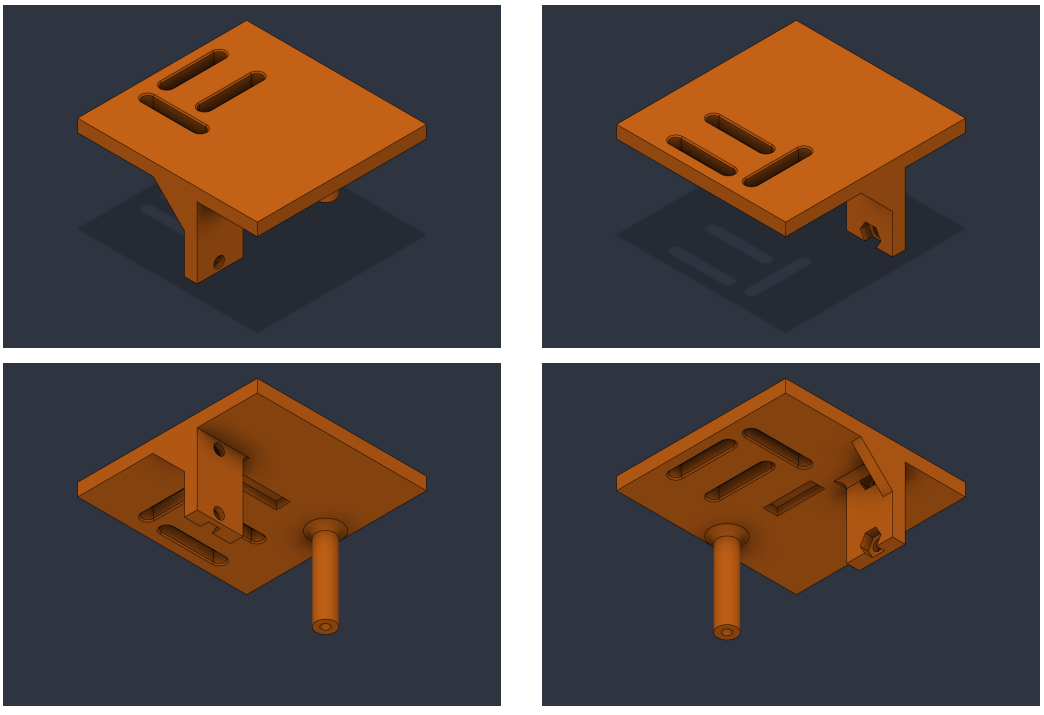


Figure 1.7: Right bracket on its own: top and underside, from the front and the rear.

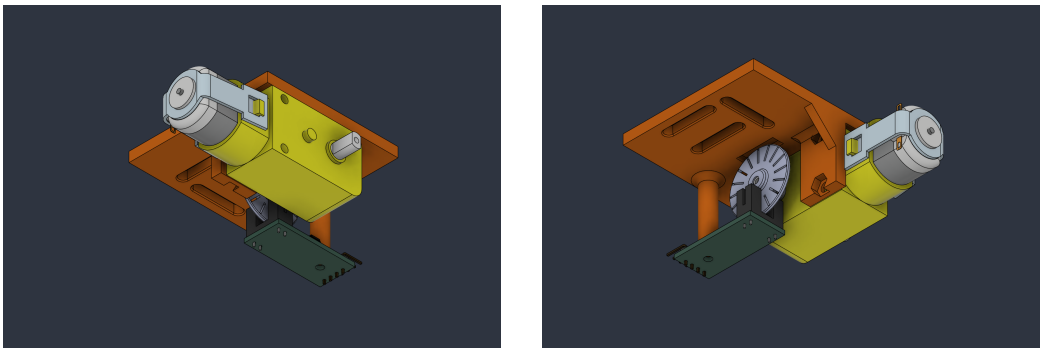


Figure 1.8: Assembly from below: motor tab and nut traps, encoder disc and LM393 sensor.

Bibliography

[1] O. Simpson, *TT Motor Dual Shaft*, CAD model (Fusion .f3d and STEP), downloaded as `tt-motor-dual-shaft-1.snapshot.1.zip`, Source of all TT motor dimensions: gearbox 22.5 x 18.8 mm, through-holes 20.6 mm behind the shaft at ±8.75 mm. TODO: add download URL. Accessed: Oct. 5, 2026.

[2] P. Pennel, *Speed sensor MH-Sensor-Series*, CAD model (STEP) of the LM393 slotted speed-sensor board and a 16-slot encoder disc, downloaded as `speed-sensor-mh-sensor-series-1.snapshot.6.zip`, Source of the board, fork and disc dimensions. TODO: add download URL. Accessed: Oct. 5, 2026.

[3] Autodesk, Inc., *Autodesk Fusion*, Computer-aided design software, Personal Use licence, Used for all CAD modelling, clearance checks and rendering, 2026.